Table 1: L2 (TDOST classes), all classes — comparable to TDOST / DeepCASAS; $m = 1$ labelled windows per cluster. W/M = weighted/macro F1.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W-F1	M-F1	W-F1	M-F1	W-F1	M-F1	W-F1	M-F1
$K = 20$ *clusters*									
k-means	Sensor hist.	0.71	0.37	0.81	0.45	0.70	0.35	0.74	0.39
k-means	Sensor+ToD hist.	0.69	0.28	0.76	0.29	0.81	0.36	0.76	0.31
k-means	MLM	0.74	0.32	**0.89**	**0.51**	0.72	0.27	0.78	0.37
k-means	MLM-time	0.74	0.32	0.84	0.41	0.74	0.26	0.78	0.33
k-means	MLM + ToD	0.63	0.26	0.80	0.32	**0.86**	**0.44**	0.76	0.34
k-means	MLM-time + ToD	0.73	0.29	0.71	0.29	0.82	0.42	0.75	0.33
SCAN	MLM	0.76	0.33	0.77	0.32	0.78	0.35	0.77	0.34
SCAN	MLM-time	0.75	**0.39**	0.81	0.46	0.78	0.36	0.78	**0.40**
DEC	MLM	0.76	0.36	0.88	0.46	0.73	0.31	0.79	0.38
DEC	MLM-time	**0.76**	0.36	0.86	0.46	0.78	0.34	**0.80**	0.39
$K = 100$ *clusters*									
k-means	Sensor hist.	0.74	0.42	**0.88**	**0.47**	0.72	0.35	0.78	0.42
k-means	Sensor+ToD hist.	0.71	0.40	0.86	0.46	0.84	**0.46**	0.80	0.44
k-means	MLM	0.77	**0.46**	0.85	**0.48**	0.77	0.40	0.80	0.45
k-means	MLM-time	0.74	0.41	0.85	0.47	0.74	0.31	0.78	0.40
k-means	MLM + ToD	**0.79**	0.41	0.82	0.44	**0.86**	**0.46**	**0.82**	0.44
k-means	MLM-time + ToD	0.76	0.44	0.84	0.46	**0.86**	**0.46**	**0.82**	**0.46**
SCAN	MLM	0.74	0.40	0.85	0.43	0.73	0.35	0.78	0.39
SCAN	MLM-time	0.76	0.39	0.83	0.43	0.78	0.41	0.79	0.41
DEC	MLM	0.75	0.37	**0.87**	0.45	0.74	0.32	0.79	0.38
DEC	MLM-time	0.76	0.37	**0.87**	0.46	0.76	0.34	0.80	0.39

Table 2: L2, rare classes masked from the average; $m = 1$ labelled windows per cluster. W/M = weighted/macro F1.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W-F1	M-F1	W-F1	M-F1	W-F1	M-F1	W-F1	M-F1
$K = 20$ *clusters*									
k-means	Sensor hist.	0.72	0.47	0.81	0.72	0.70	0.34	0.74	0.51
k-means	Sensor+ToD hist.	0.70	0.35	0.77	0.46	0.81	0.42	0.76	0.41
k-means	MLM	0.75	0.40	**0.89**	**0.81**	0.72	0.32	0.79	0.51
k-means	MLM-time	0.75	0.40	0.84	0.65	0.75	0.31	0.78	0.45
k-means	MLM + ToD	0.63	0.32	0.80	0.51	**0.86**	**0.52**	0.77	0.45
k-means	MLM-time + ToD	0.73	0.36	0.72	0.46	0.83	0.49	0.76	0.44
SCAN	MLM	0.76	0.41	0.77	0.51	0.78	0.41	0.77	0.45
SCAN	MLM-time	0.76	**0.49**	0.82	0.72	0.78	0.41	0.78	**0.54**
DEC	MLM	0.76	0.45	0.88	0.74	0.73	0.36	0.79	0.52
DEC	MLM-time	**0.77**	0.45	0.86	0.73	0.78	0.40	**0.81**	0.53
$K = 100$ *clusters*									
k-means	Sensor hist.	0.75	0.52	**0.88**	**0.76**	0.72	0.36	0.78	0.54
k-means	Sensor+ToD hist.	0.71	0.50	0.86	0.74	0.84	0.47	0.81	0.57
k-means	MLM	0.77	**0.58**	0.86	**0.76**	0.78	0.42	0.80	0.58
k-means	MLM-time	0.74	0.51	0.85	0.75	0.75	0.36	0.78	0.54
k-means	MLM + ToD	**0.79**	0.51	0.82	0.71	**0.87**	**0.54**	**0.83**	0.58
k-means	MLM-time + ToD	0.77	0.54	0.84	0.74	**0.87**	**0.54**	**0.82**	**0.61**
SCAN	MLM	0.74	0.51	0.86	0.68	0.74	0.41	0.78	0.53
SCAN	MLM-time	0.77	0.48	0.83	0.69	0.78	0.44	0.79	0.54
DEC	MLM	0.75	0.46	**0.87**	0.73	0.75	0.37	0.79	0.52
DEC	MLM-time	0.77	0.46	**0.88**	0.74	0.76	0.39	0.80	0.53

Table 3: L1 (raw CASAS activities), rare classes masked from the average; $m = 1$ labelled windows per cluster. W/M = weighted/macro F1.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W-F1	M-F1	W-F1	M-F1	W-F1	M-F1	W-F1	M-F1
$K = 20$ *clusters*									
k-means	Sensor hist.	0.70	0.46	0.82	0.63	0.18	0.20	0.57	0.43
k-means	Sensor+ToD hist.	0.69	0.36	0.76	0.35	0.67	0.35	0.71	0.35
k-means	MLM	0.73	0.38	**0.89**	**0.66**	0.21	0.20	0.61	0.41
k-means	MLM-time	0.73	0.41	0.84	0.55	0.29	0.16	0.62	0.37
k-means	MLM + ToD	0.62	0.29	0.79	0.35	**0.73**	**0.46**	**0.72**	0.37
k-means	MLM-time + ToD	0.72	0.34	0.71	0.38	0.71	0.44	**0.71**	0.38
SCAN	MLM	**0.76**	0.43	0.77	0.47	0.31	0.26	0.62	0.38
SCAN	MLM-time	0.75	**0.52**	0.82	0.60	0.27	0.23	0.61	0.45
DEC	MLM	**0.76**	0.47	0.88	0.61	0.20	0.22	0.61	0.43
DEC	MLM-time	**0.76**	0.49	0.86	0.62	0.31	0.26	0.65	**0.46**
$K = 100$ *clusters*									
k-means	Sensor hist.	0.74	0.55	**0.87**	0.63	0.27	0.24	0.63	0.48
k-means	Sensor+ToD hist.	0.70	0.50	0.86	0.61	**0.75**	**0.50**	0.77	**0.54**
k-means	MLM	0.76	**0.59**	0.85	**0.64**	0.35	0.31	0.66	0.51
k-means	MLM-time	0.73	0.53	0.86	0.63	0.31	0.28	0.63	0.48
k-means	MLM + ToD	**0.78**	0.53	0.82	0.59	0.74	0.49	**0.78**	**0.54**
k-means	MLM-time + ToD	0.75	0.54	0.84	0.62	0.73	0.44	0.77	**0.53**
SCAN	MLM	0.74	0.51	0.85	0.56	0.30	0.28	0.63	0.45
SCAN	MLM-time	0.76	0.50	0.83	0.53	0.32	0.32	0.64	0.45
DEC	MLM	0.74	0.47	**0.87**	0.60	0.30	0.26	0.64	0.44
DEC	MLM-time	0.75	0.49	**0.87**	0.62	0.29	0.27	0.64	0.46